Pedakolimi Harish

Robotics Engineer

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Summary

Robotics Engineer with **3+ years** of experience developing **ROS/ROS2-based robotic systems**, autonomous mobile robots, and robotic arm applications. Skilled in **robotics software development**, motion planning, simulation, **embedded systems**, and system integration. Experienced in bridging mechanical, electrical, and software domains to deliver real-world robotic solutions. Currently pursuing a **Master's** in Applied Mechatronics and Robotics at **IIT Bhilai** while developing next-generation robotic construction technologies.

Technical Skills

Robotics & Automation	ROS1, ROS2, Navigation, Motion Planning, Robot Kinematics, Robot Control, Autonomous Mobile Robots (AMR), Construction Robotics, Digital Twins, OpenCV, Computer Vision
Programming	C++, Python, C
Embedded Systems	STM32, ESP32, Arduino, Raspberry Pi, UART, SPI, I2C
Simulation & Design	Gazebo, NVIDIA Isaac Sim, RViz2, URDF/Xacro, Fusion 360, TF2, ros2_control, Robot Simulation, Moveit2, Nav2
Tools & Platforms	Linux, Git, Docker

Professional Experience

ROS Developer | Kelvin6k Technologies | Apr 2024 – Feb 2025

- Developed ROS2 software for **SCARA-based construction 3D printing systems**.
- Automated G-code conversation using Python, reducing preparation time by ~**20 minutes per print job**.
- Developed a robot alignment workflow that reduced setup time from **days to minutes**.
- Supported deployment, operation, and maintenance of **3 construction printing projects**.
- Collaborated across software, electronics, and mechanical teams to deliver field-ready robotic systems.

Jr. Robotics Engineer | AI BAR | Dec 2021 – Dec 2023

- Designed, built, and validated **12+ robotics and IoT prototypes** from concept to deployment.
- Developed ROS-based software for autonomous mobile robots and robotic arm platforms.
- Built **AMR, robotic arm, and IoT** systems involving software, electronics, and mechanical integration.
- Developed digital twin environments using Roblox for simulation and visualization.
- Led rapid prototyping efforts to evaluate new robotics concepts and technologies.
- Built **proof-of-concept systems** to assess technical feasibility and accelerate product development.

Robotics Mentor (Volunteer) | Sure Trust (NGO) | Nov 2024 – Present

- Mentored undergraduate students through hands-on robotics projects.
- Taught ROS, programming, electronics, and mechanical fundamentals.
- Guided students in project development and engineering best practices.

Projects

Mobile 3D Printer -

Technologies: ROS2, C++, Gazebo, Motion Planning, Localization
Project-Link

- Architected a ROS2-based mobile robotic construction platform for world-referenced fabrication workflows.
- Developed a firmware-driven motion execution system supporting trajectory streaming, fault recovery, and deterministic execution.
- Implemented coordinated mobile base and toolpath control for large-scale robotic construction applications.

SCARA Construction Printer

Technologies: ROS2, Duet 3D, Motion Control
Project-Link

- Designed and developed a SCARA robotic arm for construction-scale 3D printing.
- Implemented robot control, motion execution, and hardware integration for real-world printing operations.

Education

Master of Technology (M.Tech.) – Applied Mechatronics and Robotics
Indian Institute of Technology Bhilai (IIT), Chhattisgarh | 2024 – 2026 (Expected)

Bachelor of Technology (B.Tech.) – Mechanical Engineering
GITAM University, Visakhapatnam | 2016 – 2020